Preliminary Design Tool for Flying Wing Unmanned Aerial Systems

Malik Henry
North Carolina A&T State University

Abstract

FWT_V1-0 is a preliminary design tool for flying-wing unmanned aircraft. It keeps geometry setup, aerodynamic analysis, mission simulation, structural checks, and exports in one project model, which makes iteration faster and easier to track.

Problem and Scope

Early design work usually gets split across several tools for geometry, analysis, and manufacturing prep. This project brings those steps into one GUI workflow for flying-wing concepts so early trade studies can be done in a single environment.

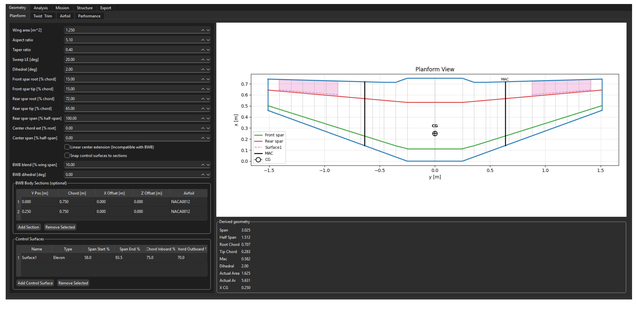
Implementation Approach

Core architecture:

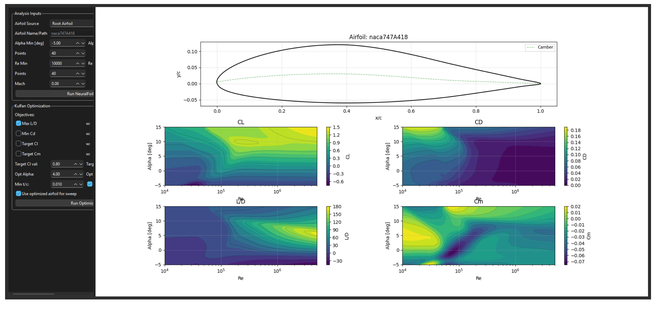
- GUI layer: Geometry, Analysis, Mission, Structure, and Export tabs.
- Shared project state: one serialized model for wing, mission, and outputs.
- Service layer: aero/geometry, mission dynamics, propulsion, structure, and export.
- Batch entrypoint: `run_full_analysis.py` for repeatable offline analysis.

Analysis pipeline currently used:

- Define planform, airfoils, control surfaces, and twist/trim targets.
- Compute performance metrics and aerodynamic polars.
- Run mission phases (ground roll, climb, cruise, descent, landing).
- Evaluate structural response with beam, stress, and buckling checks.
- Export artifacts for analysis and manufacturing (Flow5, STEP, DXF, JSBSim assets).



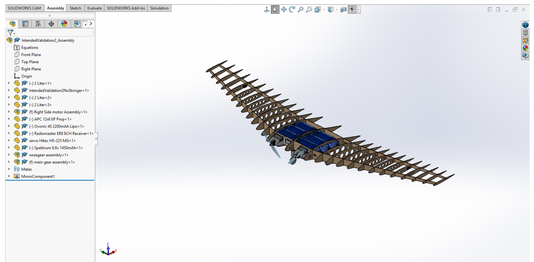
Geometry tab: planform and wing setup workflow.



Analysis tab: aerodynamic sweep and post-processing views.

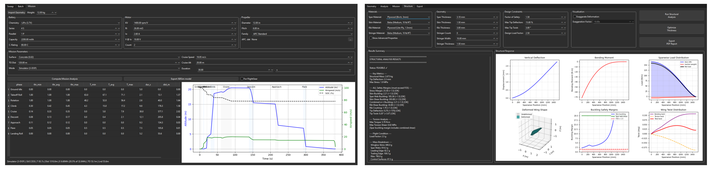
Capability Snapshot

Module	State	Notes
Geometry	Active	Planform + twist
Analysis	Partial	Airfoil opt WIP
Mission	Partial	Full mode fallback
Structure	Active	Buckling checks
Export	Partial	JSBSim debug
Batch run	Active	JSON pipeline
Scope		**End-to-end**



CAD output used for configuration and manufacturability checks.

Results



Current Gaps

- Complete the full mission-dynamics path (currently a simplified fallback path).
- Improve reliability of JSBSim export and mission autopilot edge cases.
- Add stronger regression testing for mission, structure, and propulsion outputs.
- Extend aero-structural coupling beyond the current one-way default path.

Conclusion

The current build shows a workable single-workspace flow for flying-wing preliminary design. Next work is centered on mission/export reliability and stronger verification so design-study results are easier to reproduce.

References

- Sharpe, P. D. (2021). *AeroSandbox: A Differentiable Framework for Aircraft Design Optimization*. MIT.
- Sharpe, P. D. (2024). *Accelerating Practical Engineering Design Optimization with Computational Graph Transformations*. MIT.
- Bowers, A. H., et al. (2023). *Wake-Structure of the Prandtl-D*. AIAA ARC.
- Drela, M., and Youngren, H. (1988). *AVL User Primer*. MIT AeroAstro.
- Stevens, B. L., and Lewis, F. L. *Aircraft Control and Simulation*.
- Timoshenko, S. P., and Gere, J. M. *Theory of Elastic Stability*.